Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/31/2006
PI Name:	Kimbrell, Deborah A Ph.D.		
Project Title:	Drosophila as a model of immune function in conditions of altered gravitational force		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	95616	**Congressional District:**	25
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE through 9/30/2007, per A. Chu (8/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhattacharya, Sharmila (NASA Ames Research Center)		
Grant/Contract No.:	NNA04CC76A		
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Performance Goal Text:			

Task Description:	Immune function is essential for survival and involves basic cellular processes such as non-self recognition, cell growth and apoptosis. The most fundamental part of immune function is innate immunity, and this is reflected by the fact that innate immunity is a phylogenetically ancient system. The main studies that have revealed the similarities in the immune systems of vertebrates and invertebrates are those that compare Drosophila and mammals. In particular, many components of the signal transduction cascades involved in the response to infection are conserved in flies and humans. In Drosophila, this results in the induction of antimicrobial peptides, the hallmark of the humoral response. Similarly, conserved pathways in the cellular system are involved in the regulation of immune cell growth, both for normal blood cell development and for blood cell cancer. In Drosophila, hemocytes mobilize throughout the body to phagocytose invading microbes, and also produce melanotic clusters, which are composed of melanized hemocytes that surround infecting organisms or self-tissue. The melanotic clusters involving self-tissue may either reflect a type of blood cell cancer or a benign melanotic lesion, both of which have the phenotype of melanotic tumors. This project is designed to use the power of Drosophila as a model of innate immunity to investigate the effects of gravity on immune function. A key advantage of this approach is the ability to compare normal flies to mutant flies that are either defective in an aspect of immune function or defective in gravitational response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute toward an understanding of the fundamental innate immune response and thus help identify the parameters that are important for health and targets that could be useful for disease treatment.
Task Progress:	In the past year, we have set up for experiments to analyze melanotic tumor formation in conjunction with hypergravity. These experiments will use the Gal4-UAS transgenic system of Drosophila. In this system, GAL4 specifically activates the expression of any gene that has been constructed to have a UAS promoter. For these experiments, in the last year a UAS construct of a melanotic tumor formation gene has been used to produce corresponding transgenic strains of flies. We have found that this UAS construct can restore the melanotic tumor formation phenotype of the mutant to normal. Importantly, this rescue of tumor to tumor-free requires expression only in the blood cells. We have examined parameters of immune function in Drosophila under conditions of hypergravity compared to normal gravity after infection with fungus, Beauveria bassiana, or bacteria, E. coli. Our data indicate that exposure to hypergravity changes the survival of infected flies. Specifically, flies show a significant increase ($P < 0.05$) in survival rates at 4g versus 1g. An increase in survival rate is consistent with either the infecting organism being weaker and/or the flies being stronger in response. By comparing these results to gravity sensing mutants, the relative roles of gravity, host and pathogen can begin to be discerned. We thus tested yuri mutants, named after the Russian astronaut Yuri Gagarin and isolated as having a defective gravitaxis response, for their survival after infection and exposure to 4g hypergravity. The UAS-yuri rescue stock, which restores normal gravitaxis to yuri mutants, showed the same increase in survival as the other stocks tested. The yuri mutants, however, showed no change in survival at 4g compared to 1g. Thus, the data indicate that gravity has an effect on the host response. Subsequent tests are in progress to further determine the relative effects of gravity on the functions of the host and infecting organism. In addition, we have found that the survival of infected flies is very different for male versus female flies. Generally, only male flies have been tested in immune response studies, and those that have tested females have addressed specific features, e.g. a specific immune response gene and mated/unmated conditions. Our studies have shown that females survive less well post-infection ($P<0.01$), with both fungus and bacteria, and for all wild type and mutant strains tested. Females are also more variable than males in survival rates, and less clear as to the effects of altered gravitational forces. We are currently investigating the gene expression changes post-infection for females versus males. These studies have also supported our FIT (Fungus, Immunity and Tumorigenesis) Flight Experiment, scheduled for the space shuttle flight ULF1.1.
Bibliography Type:	Description: (Last Updated: 07/07/2015)
Articles in Other Journals or Periodicals	Bernal A, Schoenfeld R, Kleinhesselink K, Kimbrell D. A. "Loss of Thor, the single 4E-BP gene of Drosophila, does not result in lethality." Drosophila Information Service, 2004 Dec;87: 81-84. , Dec-2004
Presentation	Taylor, K. A., Kloehn, A. T., Fuller, P. M., Beckingham, K. M., Fuller, C. A. and Kimbrell, D. A. "Alteration of gravitational force affects the immune response." 46th Annual Drosophila Research Conference, Houston, TX Mar-2006
Presentation	Kimbrell, D. A. "Gravity and the Immune Response." 45th Annual Drosophila Research Conference, Gravity and the Fly Workshop, Washington D. C. Mar-2004
Presentation	Kimbrell, D. A. "Tales of Drosophila Immunity." University of California Davis, Section of Molecular and Cellular Biology, Faculty Chalk Talk Series. Apr-2004